



GE Medical Systems

Technical Publications

**Direction 2209282
Revision 4**

Signa® Release 8.x Software Options Installation By MOD

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Operating Documentation

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

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- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

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1 OVERVIEW

This procedure describes how to install software options on the Signa 8.x system using the MOD drive. The Software Option package contains the appropriate software option MOD.

2 OPTION SOFTWARE INSTALLATION

1. Insert the MOD disk in the SGI computer MOD drive.
2. Select the **[Service Tools Desktop]**.
3. Select **[Install...]** from the Service Desktop Manager.
4. After Type Root Password is displayed, type **operator** as the password.

2-1 For Release 8.3

1. After the Guided Install window opens (30 sec. for Octane; 60 sec. for Indigo), select **Option Key tab**.
2. Select **MOD** box at Install Option Key portion of window, then select **Install**.
3. When installation process is complete, either load the next MOD, or when all option keys have been loaded, select **File ->Quit** to exit the Guided Install tool.

2-2 For Release 8.2.5

1. From the "Install" window, select **[Option Install]**
2. From the "Option Install" window, select **[Change Status]** next to the appropriate Option Name to be installed.
3. At the "Select Device" window, select the appropriate drive device, i.e., **[MOD]**.
4. When installation process is complete, either load the next MOD option disk and repeat steps 2 & 3, or in the "Install Keys" window, select **[Quit]**.
5. Wait about 20 seconds for various windows to clear, then from the "Option Install" window, select **[Quit]**.
6. At the "Status Changed" window, select **[OK]**.
7. At the "Install" window, select **[Quit Install]**.

3 REBOOT COMPUTER

1. Reboot the computer.
2. The installation is now complete.

4 SIGNA 8.3 OPTION CATALOG MOD CONTENTS

What follows is a list of available catalog option MOD(s) for the Signa 8.3 and later systems and a list of what software options are activated with each MOD.

EchoSpeed Option Key MOD 2209991

- Turns on the EchoSpeed Option.

M1090LD- Proactive Service

- Proactive Service - Activates Proactive Diagnostics and Dialout.

M1090LF - Phase Contrast Vascular

- Phase Contrast Vascular Imaging - CVI

M1090LG - Cine

- Cine - Images usually of moving anatomy scanned and arrange for playback rapidly in movie format.

M1090LJ - IVI

- Interactive Vascular Imaging - IVI

M1090LL - 3D Surface

- 3D Surface/Voxtool

M1090LM - Research

- Research - Allows for access to low level control of scan parameters.

M1090LN - Research PSD

- Research PSD - Advanced PSD development.

M1090LP - Echo Plus

- DW EPI - Diffusion Weighted Echo Planer Imaging

M1090LT - SmartPrep

- SmartPrep - Automated triggering for contrast enhanced MRA

M1090MW - MS/MA

- Multi-Slice/Multi-Angle Oblique

M1090NN - FuncTool

- Func Tool - Raw Data Analyses tool.

M1090PL - ConnectPro

- HIS RIS - Provides DICOM modality work list connectivity to hospital or radiology information system.

M1090KS - Lx UG Options Europe/Asia

- Spectroscopy/PROBE - Single voxel proton spectroscopy with PROBE-S and PROBE-P
- DW EPI - Diffusion Weighted Echo Planer Imaging
- Flair EPI
- SmartPrep - Automated triggering for contrast enhanced MRA
- SSFSE - Single Shot Fast Spin Echo

M1090CR - Cardiac All

- Bloodsupp - Blood Suppression
- Fastcine - Fast gated multi-phase images of the heart that show the heart as it contracts and expands during cardiac cycle in a movie format.
- Tagging - The placement of "tags" in an MR image is used to assess myocardial wall functions visually.
- Sgdperf - SGD enhancements which increases slew rates and duty cycles for higher resolution thin slice imaging.

M1033JA - iDRIVE PRO

- iDRIVE Pro Interactive imaging uses a fast GRE pule sequence and a new user interface to acquire continuous 2D images at a high frame rate and provide on-image navigation. Not available on systems with a single i860 AP.

M1033JB - SmartPrep2000

- Smart Prep - Automated triggering for contrast enhanced MRA
- SmartPrep2000 Upgrade - Provides SmartPrep automated table motion.

M1033JH - SmartPrep2000 Upgrade

- SmartPrep2000 Upgrade - Provides SmartPrep automated table motion.

M1033JD - Probe2000

- Spectroscopy/PROBE - Single voxel proton spectroscopy with PROBE-S and PROBE-P
- Probe2000 Upgrade - Multi voxel proton spectroscopy (PROBE - SI)
- Multi-Nuclear Spectroscopy - Provides access to additional spectroscopy PSD which may be used in proton mode.

M1033JJ - Probe2000 Upgrade

- Probe2000 Upgrade - Multi voxel proton spectroscopy (PROBE - SI)
- Multi-Nuclear Spectroscopy - Provides access to additional spectroscopy PSD which may be used in proton mode.

M1033JF - ScanTools2000

- Cine - Images usually of moving anatomy scanned and arrange for playback rapidly in movie format.
- Phase Contrast Vascular Imaging
- Echo Planar Imaging (EPI)
- 3D Surface/Voxtool
- Flair EPI
- iDRIVE - Basic Real Time Interactive Imaging package.

M1033JL - LX Tools (former MiniTools)

- Fast Spin Echo & FLAIR
- Fast Gradient Echo
- Time of Flight
- Interactive Vascular Imaging
- SPECIAL - Spectral / Spatial Fat Suppression. Chemical saturation using a inversion pulse (180° pulse). This key will always suppress just fat and can not be changed to suppress water.
- SSFSE - Single Shot Fast Spin Echo
- Three Plane Localizer
- E3DTOF
- FSE-XL
- Clariview - Image Filtering Package
- Sgdperf - SGD (PSD) enhancements which increases slew rates and duty cycles for higher resolution thin slice imaging.

M1033J - ScanTools2000 Upgrade

- Cine
- Phase Contrast Vascular Imaging
- Echo Planar Imaging
- 3D Surface/Voxtool
- Flair EPI
- iDRIVE Pro Interactive imaging uses a fast GRE pulse sequence and a new user interface to acquire continuous 2D images at a high frame rate and provide on-image navigation.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
3	Oct, 1999	Piepenburg	Interleaf Version
4	Feb 21, 2000	Hawthorne	Converted to Word. Added section 3 explaining what options are turned on by what MOD.